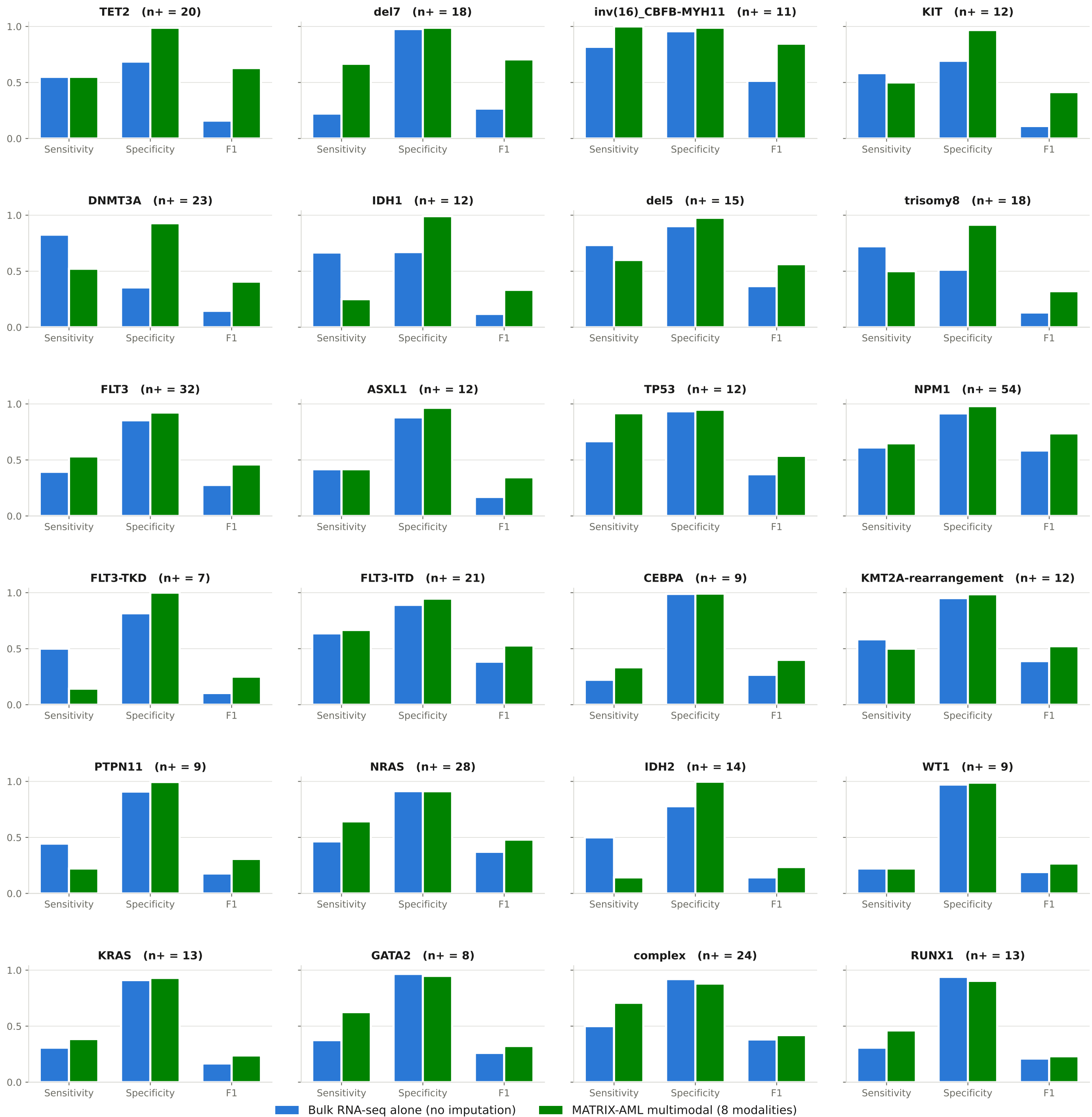


Per-mutation: sensitivity · specificity · F1

sorted by F1 gain (multimodal – bulk RNA-seq alone)



Donor-grouped 3-fold cross-validated out-of-fold predictions; both arms use identical samples, labels and folds.
 Operating point = F1-max threshold on the same OOF vector, so absolute values are optimistic — the CONTRAST between arms is the result.